Reg. No. | | | | | | | | | | | [18AS35]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER - 2021

ENGINEERING MECHANICS

Time: 03 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

1. a. Draw the Stress – Strain Relationships diagram for mild steel bar and explain each and every point on that curve. **07**

b. Obtain the equation to find out the total elongation of the tapered rod of circular cross section with D_1 and D_2 are major and minor diameters, L is length of the bar. **07**

2. a. Two plates young's modulus are E_1, E_2, and areas A_1, A_2 are joined together to get the compound bar. Obtain the equation to get the equivalent modulus of the compound bar. **07**

b. A bar subjected to a tensile force of 12 kN, if the cross section is 10 mm^2, find the Normal, tangential and resultant stress, intensity of plane making an angle of 35^0 with the plane of original stress. **07**

3. a. Draw the shear force and bending moment diagram for the 4 mts length simply supported beam subjected to 2 kN and 4 kN point loads at 1 M and 3 M distance from left support and also 4 kN load distributed uniformly over 2 mtrs length at central portion of the beam. **07**

b. Obtain the equation $M/I = \sigma/y = E/R$ from theory of simple bending to get the relation between various characteristics of a beam. **07**

OR

4. a. Draw the shear force and bending moment diagram for the 9 mts length simply supported beam subjected to a triangular load of intensity 2 kN at left support 'A' and zero kN at right support 'B'. **07**

b. A copper wire of 2 mm diameter is required to be wound around a drum. Find the minimum radius of the drum, if the stress in the wire is not to exceed 80 MPa. Take modulus of elasticity for the copper as 100 GPa. **07**

5. a. A 40 mm external diameter, 25 mm internal diameter, of an aircraft propeller hollow shaft transmits power of 600 kW at 80 rpm. The shaft is also subjected to a BM of 30 kN-m and axil thrust 250 kN. Find the principal stress and their planes **07**

b. Obtain the equations to find out the combined stiffness of two springs when they are connected in series and in parallel. **07**

OR

6.	**a.**	A solid shaft has to transmit 90 kW power at 160 rpm. Find the suitable diameter of the shaft. If the Max. Torque transmitted in each revolution exceeds the mean by 25%. Take allowable shear stress as 70 N/mm^2.	**07**
	b.	Determine the mean coil radius, wire diameter and number of turns of a closed coiled helical spring of stiffness 1 kN/m and of solid length 45 mm. The maximum shear stress in the spring under an axial load of 95 N is should not exceed 180 Mpa. Modulus of rigidity for the spring material is 80 Gpa.	**07**
7.	**a.**	Obtain the maximum deflection of simply supported beam which is not subjected to pure bending couple and the radius of curvature will not be constant at all the points.	**07**
	b.	Obtain the maximum deflection of fixed beam subjected to a point load 'P' applied at 'a' distance from left support and 'b' distance from right support. Apply energy method.	**07**

OR

8.	**a.**	Obtain the maximum deflection of simply supported beam subjected to uniformly distributed load throughout the beam by double integration method.	**07**
	b.	Obtain the maximum deflection of fixed beam subjected to uniformly distributed load throughout the beam by using Castigliano's first theorem.	**07**